TOOLBOX ARTICLE

MRFoundry: Knowledge-Guided Agent Loops for End-to-End MRI Research

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Abstract

Recent advances in AI agents create an opportunity to assist MRI research by writing code, using software, and revising computational workflows. Applying these capabilities requires connecting acquisition physics, data conventions, specialized tools, and scientific validation. MRFoundry is presented as an open-source toolbox that organizes this knowledge into seven complementary skills for existing agents. The skills connect scientific references and established tools to concrete actions and output checks. Project-local memory preserves decisions and evidence for reuse, while reviewed contributions allow researchers to improve shared guidance. Two researcher-supervised examples illustrate sequence construction and checking, acquired-data reconstruction, and the recording of unresolved questions. Guidance was consulted during a later sequence audit and during the reconstruction run. Alongside MRFoundry, MRI Agent Bench is introduced, and the guidance package is evaluated through paired offline comparisons across models and MRI task domains. MRFoundry improved aggregate verified completion on Core; evidence on harder tasks was less conclusive. The toolbox provides a practical foundation for connecting MRI knowledge, computational tools, and researcher supervision. Benefits from subsequent memory reuse remain to be evaluated.

KEYWORDS:
MRI, agentic workflows, research software, reproducibility

WORD COUNT: 3491

1 | INTRODUCTION

AI agents can write code, operate software, and use execution feedback to carry out computational work[1,2,3,4]. In scientific workflows, these capabilities offer a way to connect specialized tools, inspect intermediate results, and revise an implementation as a research question develops[5]. MRI is a natural setting for this assistance: acquisition design, simulation, reconstruction, analysis, and reporting span several software environments while sharing assumptions about the same experiment.

Effective assistance requires knowledge of acquisition physics, hardware constraints, data conventions, and computational models[6]. A reconstruction can execute successfully with incorrect axes or sampling conventions;

a valid sequence file still requires physical and numerical checks. The agent must establish what the inputs represent, preserve their meaning between applications, and select evidence that supports interpretation. For a sequence prototype, this means connecting requested timing to an exported sequence and a simulation that reads it correctly. For reconstruction, it means relating measured k-space to the sampling mask, encoding model, and comparison image. The knowledge needed for these decisions is distributed across publications, documentation, and laboratory experience. Making it accessible during a task is the opportunity addressed here.

We developed MRFoundry to organize this knowledge into seven complementary skills for existing agents[7]. Each skill links scientific references and software instructions to actions and output checks. A project-local notebook preserves the evidence behind decisions, and selected lessons can inform shared guidance through human review. The researcher supplies the question and constraints, and the agent performs computational work using established applications and supplied helper scripts. Related systems address other parts of this process: Agent4MR couples sequence generation to physics-aware validation, and BCER supplies a controller for long MRI workflows with explicit artifact binding[8,9]. MRFoundry provides reusable guidance across MRI domains, while execution and recovery remain the host's responsibility.

This paper describes the toolbox, project memory, and contribution workflow, followed by two researcher-supervised examples. The sequence example includes a later skill-guided audit; the reconstruction example documents guidance consulted during execution. Alongside MRFoundry, we introduce MRI Agent Bench to compare matched agents with and without the toolbox. Its specifications and public tasks are maintained in a separate repository[10,11]. The examples show decisions and checks in context, whereas the paired comparison evaluates completion with the frozen guidance package. Benefits from individual skills and subsequent memory reuse are not isolated by this design.

2 | METHODS

We first describe how the toolbox supplies guidance, retains project evidence, and supports reviewed contributions. We then specify the supervised demonstrations and the paired benchmark comparison, which address workflow behavior and task completion, respectively.

2.1 | Knowledge-guided research workflows

Figure 1 shows the intended loop. The researcher supplies a question, inputs, and constraints. The agent identifies relevant skills, plans actions, executes tools, and checks whether the outputs address the question. Failed checks can prompt revision of an implementation or its assumptions. MRFoundry describes this process in instructions; the host and researcher control execution and decide when further work is needed.

2.1.1 | Skill organization and MRI knowledge

Each skill is a readable instruction package with an entry point that identifies when it applies and links to focused instructions, references, and, where useful, helper scripts. Short instructions guide the next decision; linked material supplies concepts and implementation details. The host agent interprets this material in the context of the request, so the same skill can guide tasks with different data and constraints. Relevant topics can be retrieved without loading every reference. This organization separates reusable MRI guidance from the particular commands and numerical applications used in a project.

Table 1 groups the seven skills into two general and five specialist skills. The MRI navigator connects concepts, formats, software, and references across the field. The research-workflow skill connects study design, experiments, evaluation, and writing[7]. Specialist skills cover hardware, pulse-sequence design, conventional reconstruction, deep-learning reconstruction, and diffusion MRI. They address decisions such as choosing an encoding model, checking gradient conventions, or establishing reconstruction baselines. Skills can be combined when a task crosses domains: sequence design may require hardware limits and reconstruction conventions, while a learned reconstruction study requires conventional comparisons and a defensible evaluation design.

The reference index connects instructions to annotated papers, textbooks, courses, and software documentation[31]. Scientific sources explain the basis for a method, while software documentation describes its implementation. For example, a reconstruction task can begin with the specialist skill, follow a format reference to establish array conventions, and consult workflow guidance to define a baseline and endpoint. The agent and researcher can thus trace a recommended action to its source. Table 1 describes the intended coverage of the

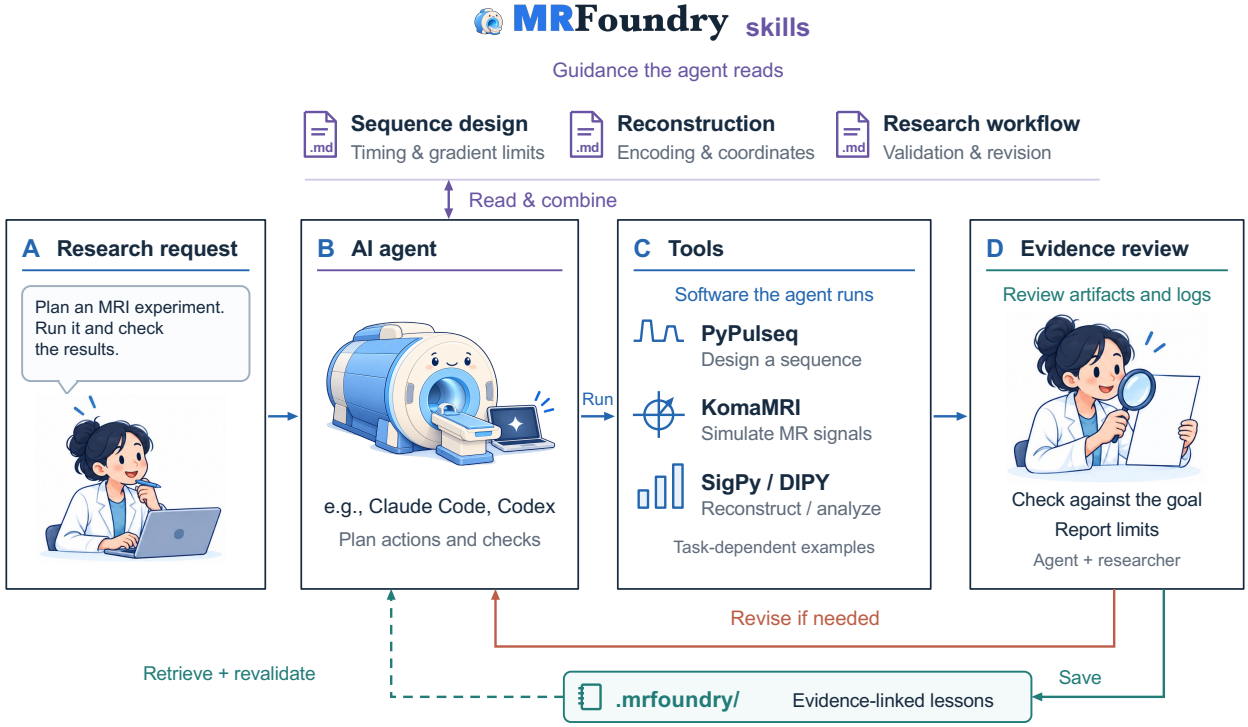


FIGURE 1 MRFoundry's guidance-based research loop[7]. A: the researcher specifies a goal. B: an AI agent consults relevant skills and plans actions and checks; Claude Code and Codex are illustrative hosts[12,13]. C: separately installed applications perform task-dependent computations, illustrated by PyPulseq, KomaMRI, SigPy and DIPY[14,15,16,17]. D: artifacts are reviewed with the researcher against the goal; insufficient evidence can prompt revision. The project-memory guide prescribes evidence-linked records, selective retrieval and revalidation[18]. Arrows describe recommended relationships rather than an enforced execution policy. Generated characters identify roles and are not scientific data. MRFoundry is compared with no skills in Table 2; the effect of project memory on later tasks was not measured.

TABLE 1 MRI-domain skills and reference entry points

MRI skill	Decisions supported	Reference entry points
MRI navigator `mri-research`	Physics, data conventions and cross-domain navigation	MRI and sequence foundations[6]
Research workflow `mri-research-workflow`	Study design, evidence review and reporting	Reproducible neuroimaging; AI reporting[19,20]
Hardware `mri-hardware`	Receive weighting and acquisition constraints	Receive-array foundations[21]
Sequence design `pulse-sequence-design`	Events, timing, trajectories and simulation	Pulseq, PyPulseq and KomaMRI[22,14,15]
Reconstruction `mri-reconstruction`	Encoding, calibration and regularization	SENSE, ESPIRiT and sparse MRI[23,24,25]; SigPy and BART[16,26]
Deep-learning reconstruction `deep-learning-recon`	Training, splits, baselines and generalization	fastMRI and end-to-end variational networks[27,28]
Diffusion MRI `diffusion-mri`	Gradients, model assumptions and fit interpretation	Diffusion tensor, DIPY and WLS[29,17,30]

The public source index links each stable skill name to its annotated references[31]. These are representative entry points. Skills provide guidance and helper scripts; broader software dependencies are installed separately. Topic coverage is not an evaluation result.

toolbox. The demonstrations illustrate selected workflows, while the benchmark assesses completion on its specified tasks.

2.1.2 | Agent workflow and tool use

Hosts such as Claude Code and Codex expose the skills to the agent[12,13]. The command `npx skills add KeWang0622/MRFoundry` installs the guidance; `--skill mri-reconstruction` selects one skill[32,7]. Scientific applications and restricted datasets retain their installation, access, and license requirements. MRFoundry combines guidance with task-specific helper scripts; broader software dependencies are installed separately.

Before execution, guidance asks the agent to inspect the input data and working environment, resolve missing information, install dependencies in isolation, and check a small upstream example[33]. When connecting applications, the agent must preserve units, axes, sample order, and model assumptions. For example, reconstruction requires distinguishing the stored grid from acquired support before constructing a mask. After execution, the agent inspects artifacts and quantitative checks against the research request. A successful command is one intermediate observation; a usable result also needs a documented input interpretation, comparison reference, and explanation of remaining uncertainty.

2.2 | Research memory and reusable lessons

The workflow also produces information that is specific to the project. Figure 2 shows how the `.mrfoundry/` notebook retains this evidence separately from reusable public skills. It contains an index, working preferences, environment information, evidence-linked lessons, and individual run records[18]. Run records retain requests, input identifiers, versions, parameters, commands, outputs, and unsuccessful attempts. Together, these details allow a later session to recover both a decision and the reasoning behind it.

A reusable lesson states an observation, its supporting evidence, its scope, and the conditions for revalidation. The intended retrieval process begins with the index: the agent selects relevant lessons and checks their applicability to the current data and environment. It may retain, narrow, or reject a lesson, with conflicting evidence superseding an earlier interpretation while preserving the original record. User preferences, such as a reporting format, are stored separately from scientific findings so that a local working habit does not become a methodological rule.

This proposed recursive self-improvement cycle operates on written guidance rather than model weights. The initializer can add retrieval instructions to project `AGENTS.md` or `CLAUDE.md` files; the host controls their loading and execution[18]. A recorded command may depend on a particular software version or dataset, making revalidation part of reuse. The notebook therefore retains both the lesson and the evidence needed to judge whether it still applies. Our demonstration covers evidence-linked recording. Later retrieval, application, and effects on task performance were not evaluated.

2.3 | Local adaptation and public contribution

Project memory also provides a starting point for improving shared guidance. Researchers can keep a lesson local, adapt instructions to group conventions, or propose a generalizable change to the public toolbox. Project notebooks are excluded from Git by default. Preparing a public contribution requires selecting the relevant instruction, removing private context, and supplying a scientific source or tested example[18,34]. The notebook itself is not automatically published.

Local adaptation is useful when a group has established data formats, software environments, or reporting conventions. The adaptation should distinguish project requirements from scientific assumptions that need verification. For example, a preferred folder structure can be adopted as a working convention, whereas a sampling rule must be justified for the data being analyzed. Making this distinction explicit allows a group to tailor its workflow without presenting a local practice as general MRI guidance. The project record retains the data-specific context for later revalidation. A proposed public contribution should explain the problem it addresses, the conditions under which it applies, and the evidence supporting the recommended action[18,34].

Contributors edit a skill or reference on a branch and open a pull request (PR). Automated checks validate repository structure and links; a maintainer reviews scientific relevance and scope before merging[34]. A group can then review the update before adopting it. For example, an observation about one dataset's array dimensions can motivate a general instruction to verify the release format, while the particular dimensions remain in the local record. This process gives a broader instruction a traceable source without treating every project observation as universally applicable. The contribution mechanism is implemented, but its effect on scientific quality or performance was not evaluated.

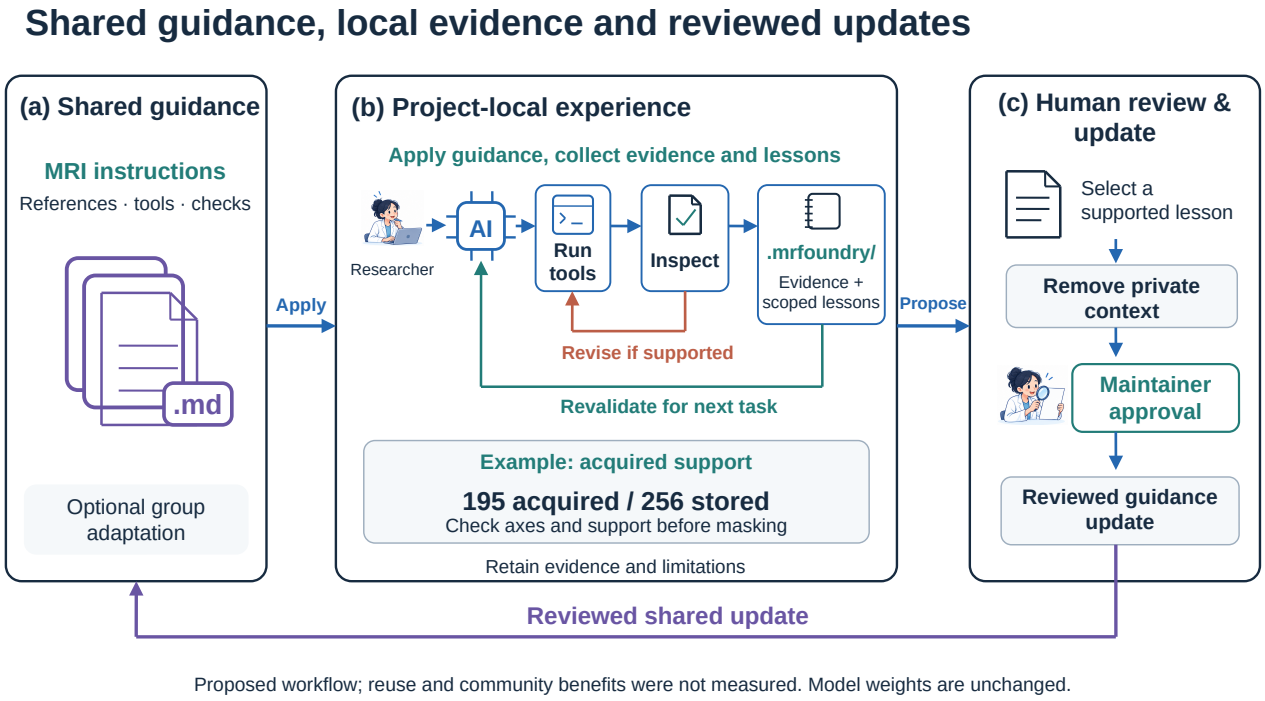


FIGURE 2 Shared guidance, local experience and reviewed improvement. (a) Public MRFoundry skills provide reusable MRI instructions and source links; a group can adapt these instructions to its methods and environment. (b) Project-local records retain preferences, evidence and scoped lessons. The researcher and agent execute tasks, inspect outputs and record evidence; current work can be revised when supported, while retrieved lessons must be revalidated for a new task. The acquired-support example distinguishes 195 acquired from 256 stored phase-encoding columns in the inspected source; the value is not a general acquisition rule. (c) Researchers select supported changes, remove private context and supply evidence for maintainer review before shared guidance is updated. Arrows describe the proposed guidance workflow rather than an enforced execution policy. This improvement loop was not evaluated in the fixed-snapshot Core/Hard comparison; model weights are unchanged.

2.4 | Demonstration design

We used two researcher-supervised Codex examples to examine how agent actions, tool outputs, and scientific checks fit together (Figures 3-4). Saved scripts, reports, and project records document the requests, tool use, and revisions. Sequence guidance was retrieved during a later audit rather than original construction; reconstruction guidance was read during the run. Neither example had a matched unguided baseline, so they illustrate workflow behavior rather than estimate the effect of guidance. Supporting Information retains numerical details, provenance, implementation history, and a separate diffusion example. NumPy, SciPy, and Matplotlib supported computation and display[35,36,37].

2.4.1 | Designing and checking a sequence

The task was to design, export, and inspect a golden-angle radial gradient-echo (GRE) prototype. PyPulseq 1.5.0.post1 produced a Pulseq file with a 220-mm field of view, 5-mm slice, 10-degree nominal flip, TE/TR 5/15 ms, and 128 spokes of 64 samples[14,22]. Successive spokes rotated by 111.246 degrees[38]. Configured per-axis gradient and slew limits were 32 mT/m and 130 T/m/s.

The agent constructed RF, gradient, and ADC events, checked timing and gradients, and compared trajectories before and after export. Readback tested whether the file preserved the coordinates that another application would receive. At the researcher's request, the agent replaced an initial custom simulator with KomaMRI[15]. It inspected the installed API and ran the official EPI example to check the environment before simulating the exported GRE on a 13,497-spin extruded 2D brain phantom. It then halved the RF integration step to assess temporal discretization on the same phantom. This sequence of checks connected the requested acquisition to its exported representation and simulated signal, with each step addressing a distinct potential implementation error.

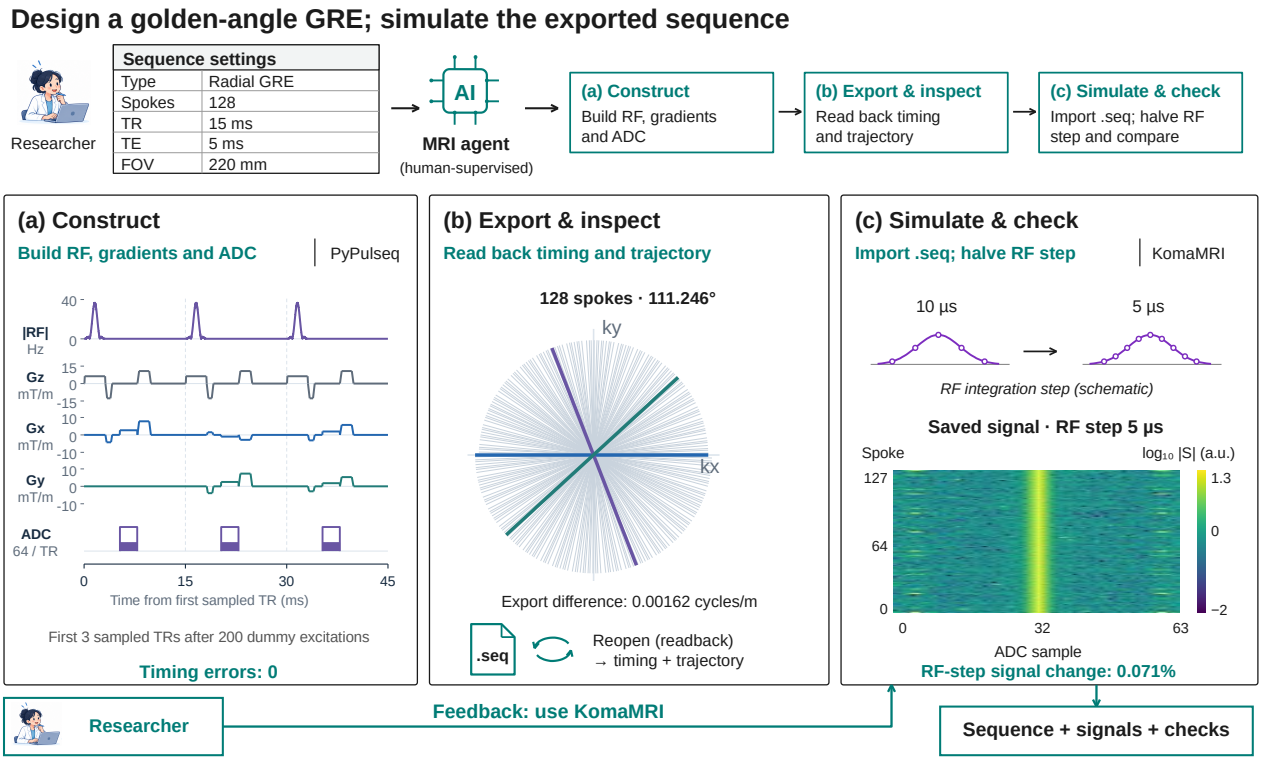


FIGURE 3 Researcher-supervised sequence construction followed by a skill-guided audit. The setup table combines agreed settings and the recorded 128-spoke implementation. (a) PyPulseq constructs RF, gradient and ADC events[14,22]. Native waveforms show the first three sampled repetitions after 200 dummy excitations; RF magnitude omits phase. Each repetition contains 64 ADC samples. Readback found no timing errors. (b) Exported spokes rotate by 111.246 degrees[38]. The unscaled geometry has nominal radius approximately 145.45 cycles/m; export/prewrite coordinates differ by at most 0.00162 cycles/m. (c) At the researcher's request, the agent replaced an initial custom simulator with KomaMRI, inspected its installed API, ran its official EPI example and simulated the exported GRE[15]. RF-step curves are sampling schematics. The heatmap displays log10 magnitude of the saved 5-μs-step signal, with zero-based spoke/ADC indices and the stated display range. Halving the RF step from 10 to 5 μs changed the complete complex signal by 0.071% (relative L2 change; exact value and definition in Supporting Information S2.1). Action labels summarize records, not a transcript. Construction and simulation preceded the documented skill retrieval at audit time. These checks do not establish spatial convergence, image fidelity, SAR/PNS compliance or scanner readiness.

2.4.2 | Comparing reconstructions

The task was to reconstruct an acquired four-channel, 0.3-T T2-weighted M4Raw brain slice at two retrospective sampling settings[39,40]. The agent consulted guidance, inspected the HDF5 array with h5py[41], and used the release contract to resolve axes and acquired support[42]. It first ran SENSE and inspected image discrepancies and measurement residuals[23]. After checking calibration, it retained the ESPIRiT maps and tried L1-wavelet reconstruction with weight 0.01 at both masks[16,43,24,25,44].

The comparison preserved initial SENSE outputs and held sensitivity maps and reconstruction settings fixed between masks. We examined image discrepancy, data agreement, and numerical stability separately. Magnitude discrepancies used the noisy all-source root-sum-of-squares (RSS) image as reference. Sampled-line residuals assessed agreement with measurements used in the solve; withheld-line residuals used acquired measurements omitted from it. Additional iterations tested numerical stability. All images used the same source-derived normalization and display window. Together, these checks distinguish an optimization output that has stabilized from a reconstruction that agrees with the measured data.

2.5 | MRI Agent Bench and comparison

MRI Agent Bench complements the demonstrations with executable research tasks, specified inputs and artifacts, and automated checks (Figure 5). Public Core contains 24 task types across ten MRI domains and 88 data variants[10,11]. Coverage includes sequence design, reconstruction, quantitative imaging, analysis,

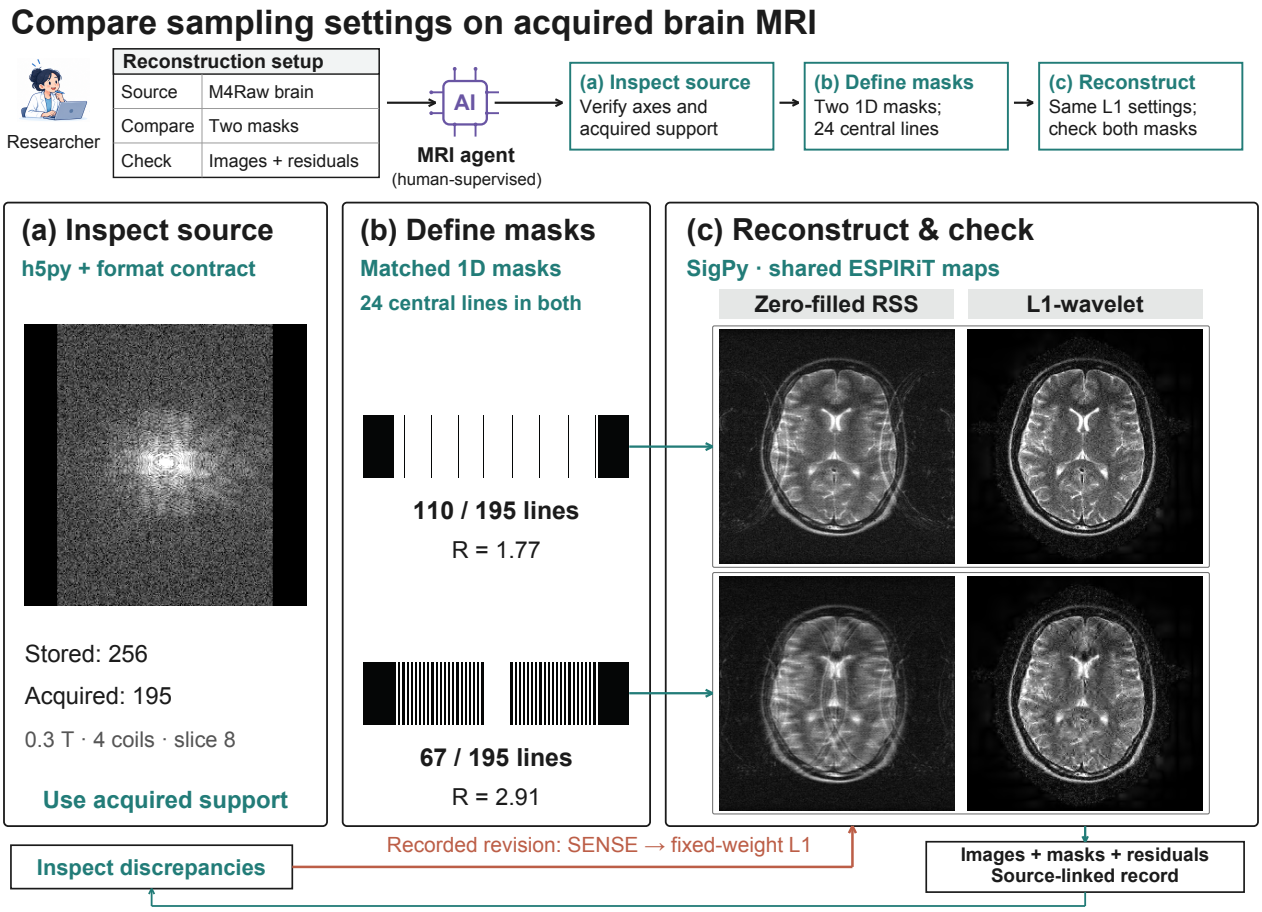


FIGURE 4 Recorded reconstruction workflow on one four-coil, 0.3-T M4Raw T2-weighted brain slice[39,40]. (a) The release contract identifies the final array axis as phase encoding; the source has 195 acquired lines in a 256-column grid. Grayscale k-space shows log10 root-sum-of-squared coil magnitudes, with fixed bounds −1.306 to −0.223 (1st-99.5th percentiles of nonzero acquired values; zero padding is black). (b) Exact retrospective masks retain 110 or 67 acquired lines, including 24 central lines, giving effective accelerations of 1.77 and 2.91. Each mask connects to a framed row containing both methods. (c) Saved zero-filled RSS and L1-wavelet images share source-RSS normalization and a window of 0-0.676 of the source RSS maximum, without per-image rescaling or inferred anatomical flips. Both L1 conditions use shared ESPIRiT maps, db4 wavelets, weight 0.01 and 200 iterations[16,24,25]. The agent revised initial SENSE to fixed-weight L1 after inspecting disagreement. Zero-filled RSS remained closer to the noisy reference, and substantial data residuals remained unresolved. Supporting Information retains the initial results and complete diagnostics. Action labels summarize records, not a transcript; the setup table combines agreed settings and recorded choices. The saved brain images retain their original pixels and common window. This supervised example is not a controlled agent-performance comparison.

and research planning and reporting. Hard comprises 11 multi-stage task types and four broader system-design types. A type specifies a problem structure; an instance fixes data and parameters. Detailed specifications and submission instructions are maintained in the accompanying repository. Separate author agents without access to the skills wrote specifications and private graders and generated data, parameters, and wording. Reference solutions and positive and negative controls checked instances before hashing and sealing. These checks assessed automated behavior; independent external MRI-expert review remains incomplete.

We compared paired runs with and without MRFoundry. Each pair ran concurrently through Codex in separate sandboxes with the same model, nominal reasoning setting, host tools, inputs, and time limit[13]. Search was disabled; network access was limited to package indexes. The intervention comprised seven frozen skill packages, including helper scripts, plus their project-level discovery instruction; the agent chose whether to read them. The control omitted this package while retaining the same common host system-skill catalog. The snapshot was fixed before evaluated Core instances were opened; subsequently authored Hard tasks were not used to revise it[45].

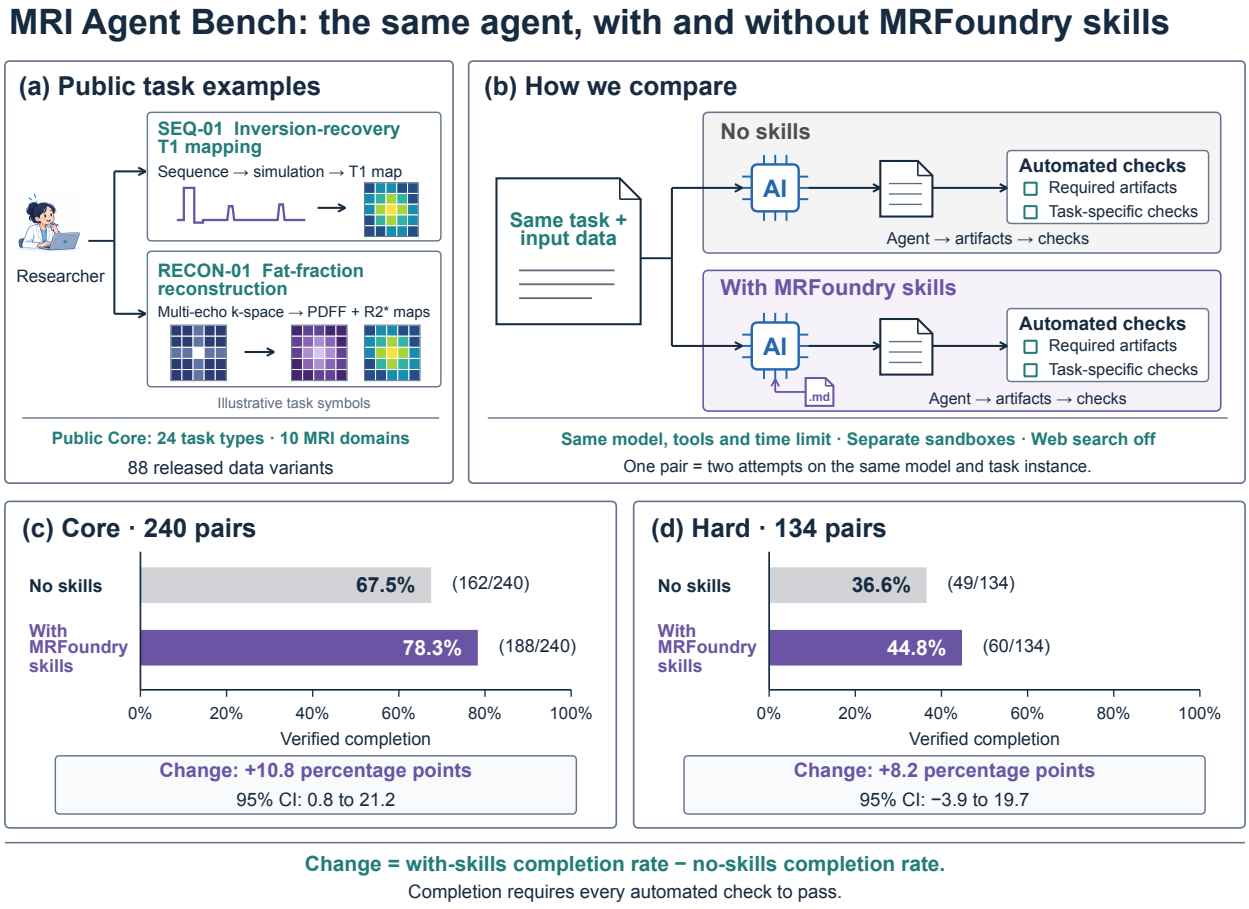


FIGURE 5 MRI Agent Bench comparison. "No skills" excludes MRFoundry and its discovery instruction; common host system-skill catalogs were matched. A: public Core examples require inversion-recovery sequence simulation, reconstruction and T1 fitting (SEQ-01), or PDFF/R2* reconstruction from multi-coil, multi-echo k-space (RECON-01)[10]. Icons are illustrative; these tasks differ from Figures 3-4. B: paired attempts use the same model and task instance in separate sandboxes, with matched inputs, host tools and time limit. The intervention is seven frozen skill packages, including helper scripts, plus their discovery instruction; skill reading is agent-selected. Web search is disabled, with package-index access allowed. Each arm produces artifacts assessed by the same checks; empty checklist symbols do not denote success. C-D: saved all-checks-passed rates on common 0-100% scales. Change is with minus without skills, in percentage points; boxes report the saved two-way task/model cluster-bootstrap 95% intervals. Core covers 24 task types × ten models (240 pairs). Hard combines three models × 15 types × two instances (90 pairs) and four models × 11 multi-stage types × one instance (44 pairs). Its interval includes zero, and unequal coverage precludes a common-task model ranking. Supporting Information gives resampling details, secondary unclustered tests and domain/cohort breakdowns.

Ten models each attempted one sealed instance per Core type, giving 240 pairs rather than evaluating all 88 variants. For Hard, three models attempted two instances of all 15 types (90 pairs); four further models attempted one instance of the 11 multi-stage types (44 pairs; Table 2). Verified completion required every frozen automated check to pass. Checks examined submitted artifacts independently of the agent's report, including exported sequences, numerical outputs, code, and claims against supplied results. No language-model judge supplied completion labels.

The primary effect was the difference in paired completion proportions. A two-way task-type/model cluster bootstrap supplied 95% percentile intervals (4,000 resamples, seed 0), interpreted as sensitivity to the selected panel. Supporting Information gives the original algorithm, secondary unclustered McNemar comparisons, task validation, grading, and budgets.

All model-reaching trials for the ten analyzed models were retained. Nine started pairs for an eleventh model were excluded after its API rejected the host's editing tool. We restarted 143 pair setups (94 Core, 49 Hard) that failed before either model turn, keeping task/model conditions unchanged. Memory and disk policies changed during evaluation. Documented memory limits were matched within newly prepared pairs, but records do not establish uniform resources throughout the study. Supporting Information documents policy

TABLE 2 MRI Agent Bench comparison by model and evaluation cohort

Model	Core: 24/model		Hard: 30/model		Hard: 11/model	
	No skills	With MRFoundry	No skills	With MRFoundry	No skills	With MRFoundry
Claude Haiku 5.5	15	19	-	-	2	3
Claude Opus 5.5	22	22	16	20	-	-
Claude Sonnet 5.5	17	23	-	-	-	-
Gemini 3.8 Flash	19	22	-	-	2	4
Gemini 4 Argon	21	24	13	14	-	-
GPT-5.3 Codex	9	13	-	-	1	1
GPT-6.1 Sol	19	22	15	18	-	-
GPT-6.1 Terra	22	22	-	-	-	-
Kimi K3	13	16	-	-	-	-
MAI Code 1.1 Flash	5	5	-	-	0	0
Completed	**162**	**188**	**44**	**52**	**5**	**8**
Paired trials	240		90		44	
Discordant pairs	39 / 13		16 / 8		4 / 1	

MRFoundry denotes the frozen evaluated snapshot (Supporting Information). "No skills" excludes MRFoundry and its discovery instruction; common host system-skill catalogs were matched. Completion requires all checks. Models are alphabetical; a dash denotes no evaluation. Core: one instance per type (24 types/model). Hard 30/model: two instances of 15 types; Hard 11/model: one instance of 11 multi-stage types. Discordant pairs: with-MRFoundry-only / control-only completions. Unequal Hard coverage precludes common-task model ranking. Pooled Hard: 49 to 60/134; system-design subset: 1 to 3/24. Figure 5 reports aggregate effect intervals.

changes, missing exposure records, and a retained Hard image-subcheck weakness.

3 | RESULTS

3.1 | Sequence and simulation checks

Figure 3 follows the research request through sequence construction, simulation, and inspection. Readback found no timing errors; maximum per-axis gradient and slew were 12.735 mT/m and 125.266 T/m/s, within configured limits. The 128 spokes contained 8,192 ADC samples. Exported/prewrite coordinates differed by at most 0.00162 cycles/m, and halving the RF step changed the complex signal by 0.07096%.

The resulting Pulseq file, trajectory, and simulation records made these checks inspectable at each connection between tools. They support timing, trajectory transfer, and one temporal-discretization choice. They do not establish full physical-model convergence, scanner readiness, or diagnostic image quality. The researcher's intervention in simulator selection remains part of the demonstrated workflow.

3.2 | Sampling and reconstruction comparison

Source inspection found 195 acquired phase-encoding lines in the 256-column grid. Masks retained 110 and 67 acquired lines, giving effective accelerations of 1.77 and 2.91 (Figure 4). Outputs included complex reconstructions, masks, sensitivity maps, and source-linked summaries. All-source RSS matched the provider's reference with relative L2 error 8.85×10^{-8}, verifying source-image construction rather than undersampled reconstruction accuracy.

L1-wavelet reconstruction reduced magnitude discrepancy relative to the initial SENSE outputs, but zero-filled RSS remained closer to noisy all-source RSS at both settings. Substantial sampled- and withheld-line residuals persisted despite small changes with additional iterations. The saved comparison therefore documents an inspected revision with unresolved data agreement. Supporting Information retains complete diagnostics; the noisy reference does not establish image accuracy.

3.3 | An evidence-linked research lesson

The reconstruction example also produced a project-memory record linking the 195-line denominator to its source digest, release contract, and numerical report (Figure 2). The reusable lesson is to verify axes and

acquired support before masking or calculating acceleration; the value 195 remains specific to this source. The initial calculation already used that denominator, so the record documents the basis for a decision rather than correction of an executed 256-based estimate. It shows how evidence and scope can be retained for later revalidation. Performance on a subsequent task was not measured.

3.4 | MRI Agent Bench comparison

Across 240 Core pairs, runs with MRFoundry achieved 188 completions compared with 162 without it (78.3% versus 67.5%; difference 10.8 percentage points, 95% CI 0.8 to 21.2). Of the discordant pairs, 39 completed only with MRFoundry and 13 only without it (Figure 5). The aggregate gain therefore includes both improvements and regressions on individual pairs. Seven models improved in aggregate Core count and three were unchanged, although equal totals could contain different successful pairs (Table 2).

Across 134 Hard pairs, completion increased from 49 to 60 (36.6% to 44.8%; difference 8.2 points, 95% CI −3.9 to 19.7), with 20 completions only with MRFoundry and nine only without it. The interval includes zero. Table 2 separates the Hard cohorts because task coverage differs; these results do not support a general model ranking. Supporting Information retains Core domain/model results, Hard cohorts, and token-use measures. Both comparisons assess availability of the complete guidance package under the recorded execution conditions.

4 | DISCUSSION

4.1 | Connecting MRI knowledge to agent actions

MRFoundry addresses the practical gap between executing an MRI workflow and establishing a basis for interpreting its outputs. Its contribution is to make domain knowledge available when an agent selects an action or evaluates a result. Because the guidance is separate from both the host and numerical software, researchers can adopt it within an existing environment and inspect the instructions that shape the work. This is particularly useful for conventions that otherwise remain implicit, such as acquired support or the meaning of a reference image. An instruction can be traced to a source, revised when its assumptions change, and reviewed before wider use.

The Core comparison provides evidence for the usefulness of making this guidance available under the tested conditions. Hard evidence was less conclusive, and pair-level gains coexisted with regressions. The comparison evaluates the package as a whole: all seven skills and their discovery instruction were supplied together, with reading left to the agent. It therefore cannot identify which instructions, helper scripts, or retrieval decisions caused an outcome to change. Understanding those mechanisms will require component-level comparisons and records of how guidance is used.

4.2 | Researcher supervision and memory reuse

The demonstrations clarify the role of researcher supervision. In the sequence example, the researcher directed the agent to an established simulator; the later skill-guided audit provided a separate opportunity to inspect the workflow. In reconstruction, stable iterative outputs still showed unresolved measurement residuals, and zero-filled RSS remained closer to the noisy reference. These observations show why inspecting an intermediate result can be as important as producing it. The examples document decisions, revisions, and remaining questions within supervised research practice; they do not establish autonomous discovery or consistently superior reconstruction.

Project memory is intended to carry this account into later sessions. Self-Refine studies revision through textual feedback, and Reflexion retains feedback across attempts[46,47]. MRFoundry records a lesson together with the evidence that defines its scope. The acquired-support example illustrates this recording step, including why a source-specific value must be rechecked. Establishing a benefit from recursive self-improvement will require longitudinal comparisons of lesson retrieval, application, and subsequent outcomes. The present study did not isolate effects of memory reuse or community contributions.

4.3 | Evaluation scope and future work

The benchmark findings have limits related to both task design and execution. Tasks were agent-authored within an author-selected scope; Core shares problem structure with development tasks, and independent expert review is incomplete. Automated checks measure specified outputs, and a retained Hard image-subcheck weakness limits the corresponding completion labels. They do not measure researcher effort, scientific judgment, or clinical validity. Independent MRI task review and stronger validation of checks are therefore priorities for extending the resource.

One run per Core task-model combination does not establish repeated-run reliability. Hard coverage differs, and nominal reasoning settings do not equalize provider budgets. Historical model snapshots, full execution configurations, and per-run resource exposure are unavailable; setup incidents and shared resources further limit interpretation. Future comparisons should include repeated runs and complete execution records, and test transfer to new acquisition conditions and host environments. Broader studies should also evaluate whether reviewed contributions improve subsequent work across projects. MRI Agent Bench provides an accompanying resource for task-completion studies, while research efficiency, scanner safety, and diagnostic benefit require their own evaluation designs.

5 | CONCLUSION

MRFoundry connects MRI knowledge, established software, and researcher supervision through reusable skills and evidence-linked project records. The demonstrations make agent decisions and checks inspectable while retaining unresolved reconstruction findings. Alongside the toolbox, MRI Agent Bench provides paired assessment of verified task completion. Completion improved on Core under the tested conditions, whereas evidence on Hard was less conclusive. These findings support continued development of reusable MRI guidance; benefits from subsequent memory reuse and shared contributions remain to be evaluated.

6 | DATA AND SOFTWARE AVAILABILITY

MRFoundry source and contribution guidance are available at https://github.com/KeWang0622/MRFoundry[7,34], with the evaluated snapshot preserved separately[45]. The reference index and tutorial provide entry points[31,48]. MRI Agent Bench publishes Core tasks, data, checkers, and result exports[10,11]. Hard materials, per-task outcomes, and traces remain private; only aggregates are reported. The paper evidence directory supplies original demonstration scripts, reports, input identities, recorded environments, and historical evaluation settings, with gaps and reproduction limits stated explicitly. Raw data remain in their original releases; M4Raw Brain v1.6 is CC BY 4.0[40].

7 | AI ASSISTANCE DISCLOSURE

AI tools supported software development, writing, and explanatory illustrations under the author's direction. Scientific MRI images and quantitative plots derive from recorded data and numerical outputs without generative enhancement. The author is responsible for the scientific content and final manuscript.

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